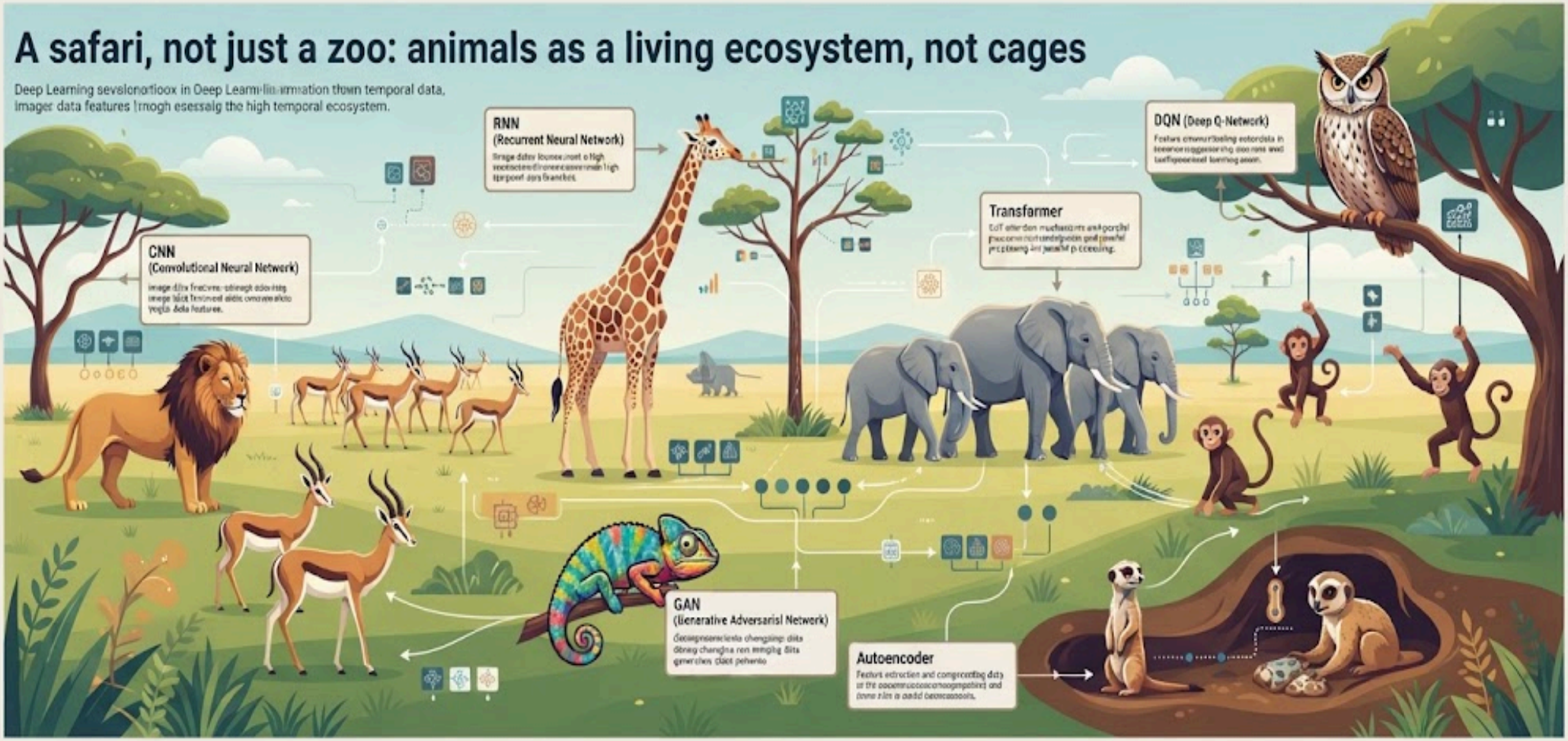
A03: The Neural Network Zoo

Team 3 | ITAI 2376 - Deep Learning
Submitter: Michael Rivas

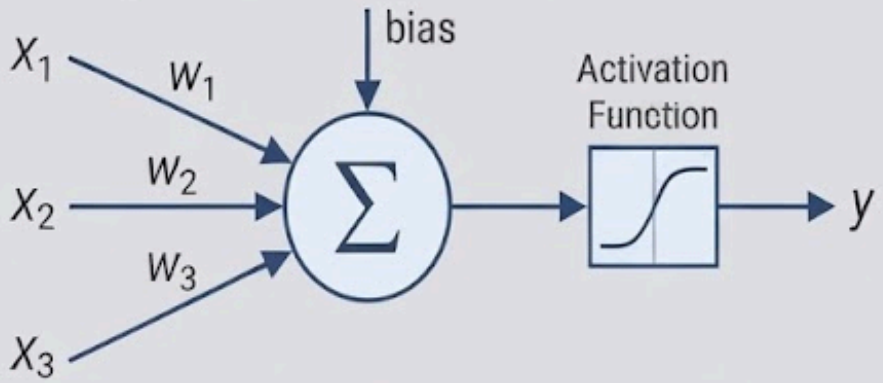


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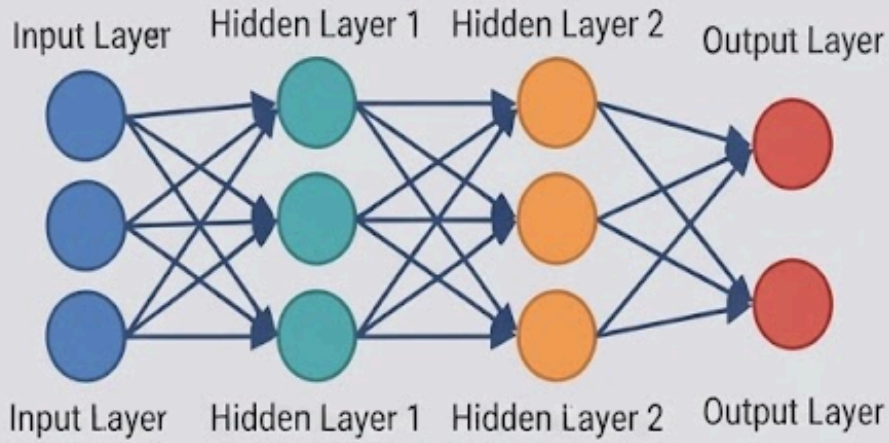
1. What is a neural network?

- Neural networks are computing systems inspired by biological brains.
- They process inputs by applying weights and adding a bias, followed by a non-linear activation function.
- Activation functions (e.g., ReLU) determine which signals are passed forward.

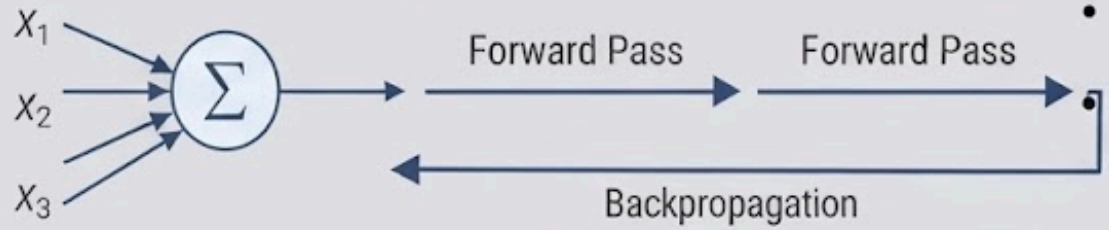
1. Neurons weigh inputs, add a bias, and fire through an activation function



2. Layers: input -> hidden layers -> output



3. Learning process: forward pass and backpropagation



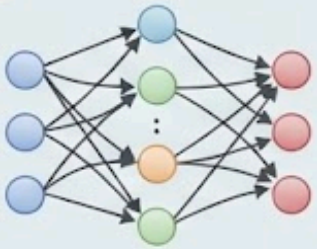
- Forward Pass: Data propagates from input to output layer to generate a prediction.
- Loss Calculation: Predicted output is compared to actual target value to calculate error.
- Backpropagation: The network calculates error gradients and adjusts weights backwards to minimize future loss.

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2. The Zoo / Safari Map

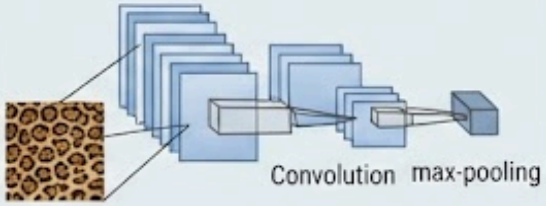


Ostrich - FNN / MLP: straight ahead, no loops



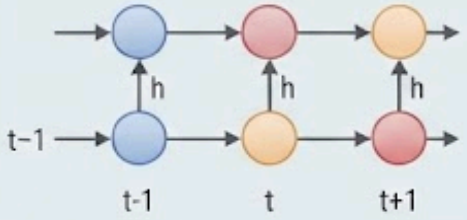


Leopard - CNN: local patterns, like rosettes, become objects



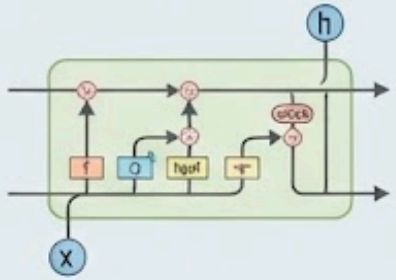


Hyena - RNN: calls relayed step by step, hidden state carried forward



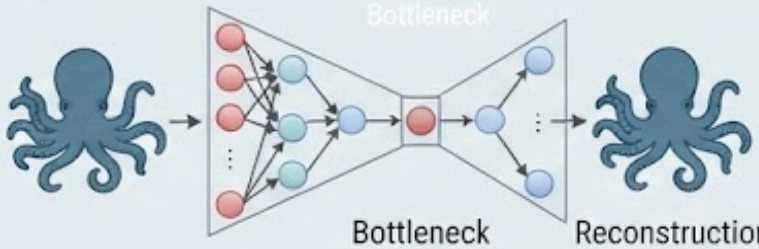


Parrot - LSTM: gates remember and forget over long sequences



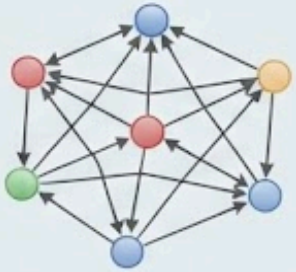


Octopus - Autoencoder: squeeze through a bottleneck, then rebuild



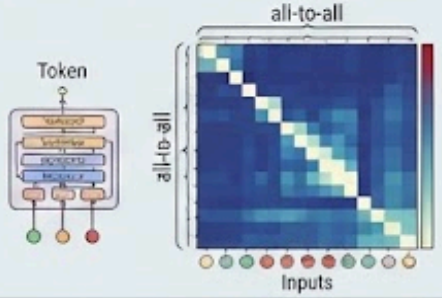


Spider - GNN: messages across a web of nodes





Honeybee - Transformer: every token can attend to every other token



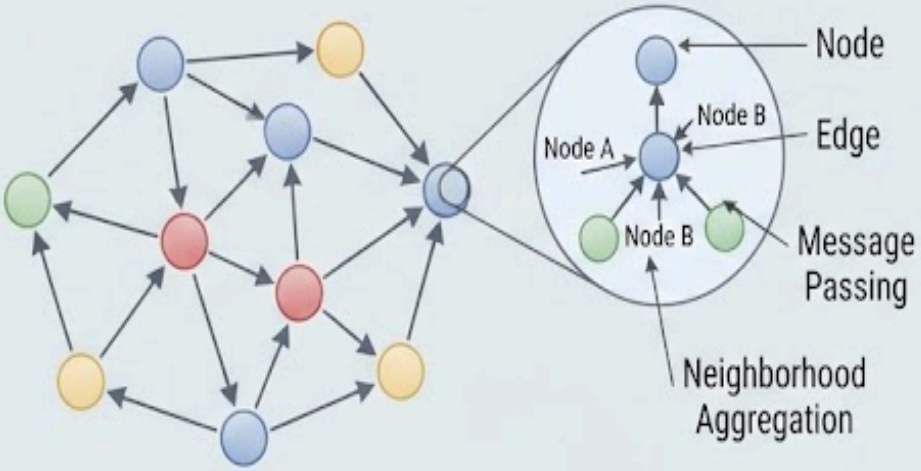
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3. Interactive Group Activity

Spider / GNN

Research & Structure: Understand Graph Neural Networks.
Focus on message passing across a web of nodes.

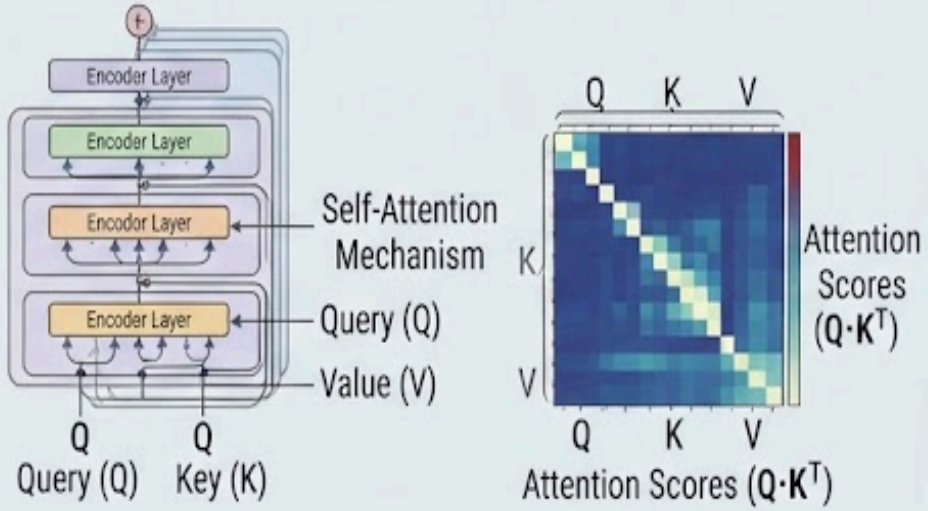


Live Notebook Cell Iteration

```
def update_node(node, neighbors):
    messages = [n.message for n in neighbors]
    node.hidden_state = aggregate(messages)
    return node.hidden_state
```

Honeybee / Transformer

Research & Structure: Understand Transformer Networks.
Focus on all-to-all token attention.



Live Notebook Cell Iteration

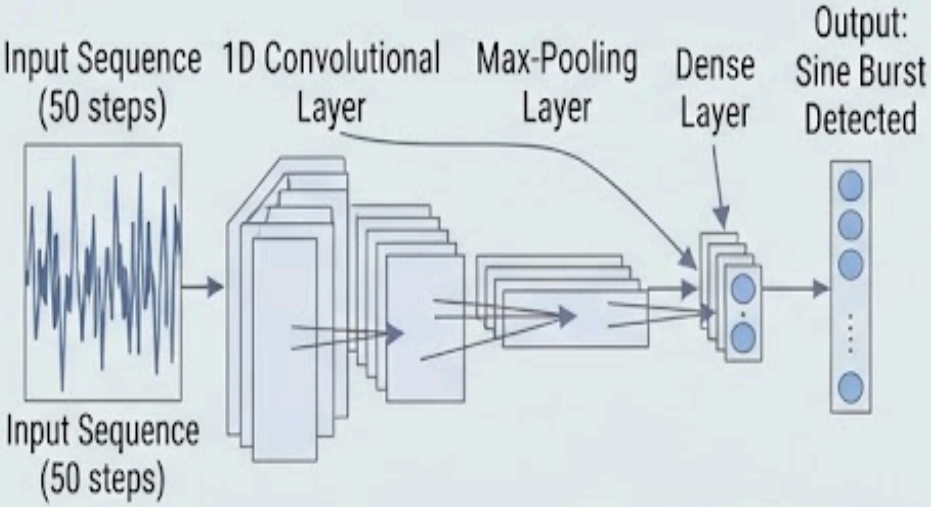
```
def scaled_dot_product_attention(Q, K, V):
    scores = matmul(Q, K.T) / sqrt(dk)
    attention = softmax(scores)
    return matmul(attention, V)
```

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4. A working program: CNN vs LSTM

Task & Data: Task: find a short sine burst in 50-step noisy sequences | **Data:** 1,000 sequences, 800 train / 200 test, 15 epochs, Adam

Leopard 1D CNN

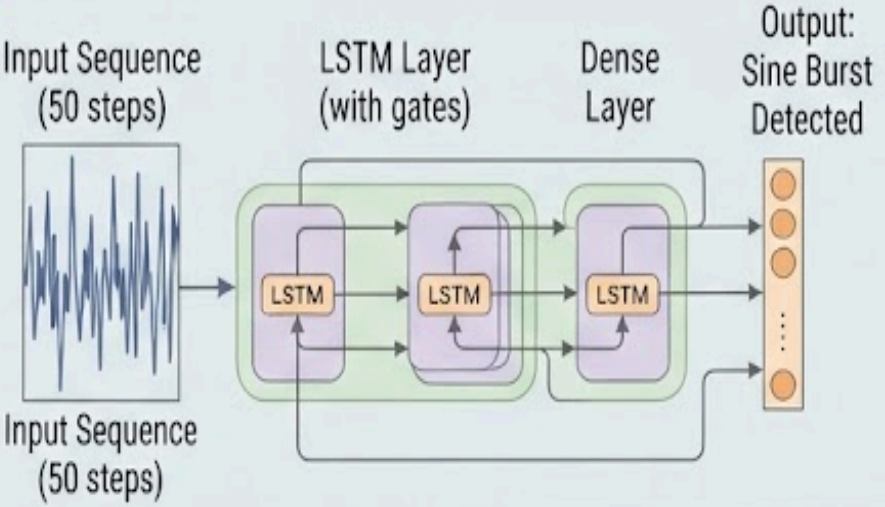


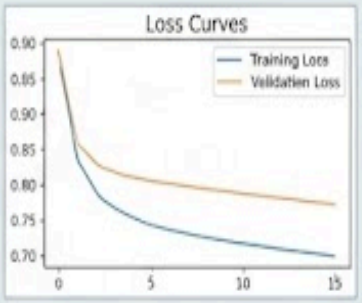


Test Accuracy: 98.00%

Loss curves are in the notebook, with both models trained on the same data

Parrot LSTM





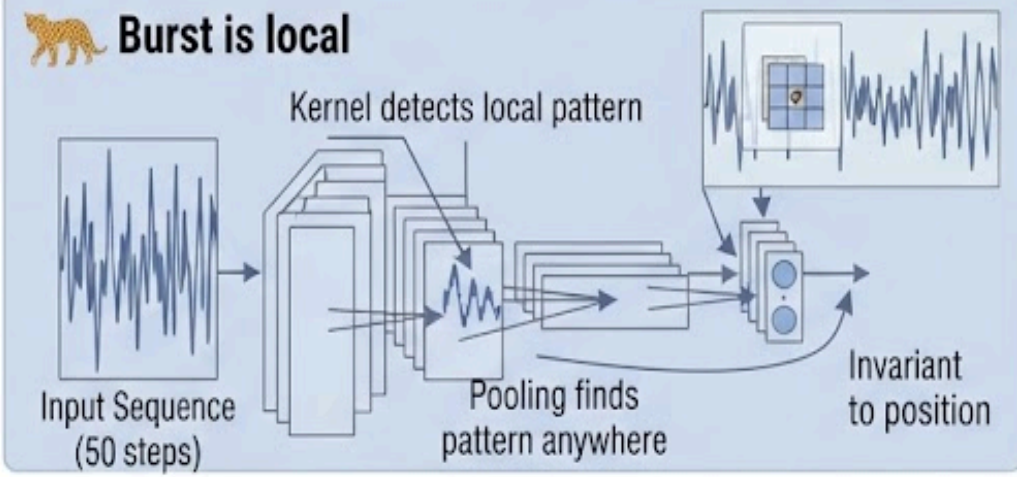
Test Accuracy: 48.50%

Loss curves are in the notebook, with both models trained on the same data

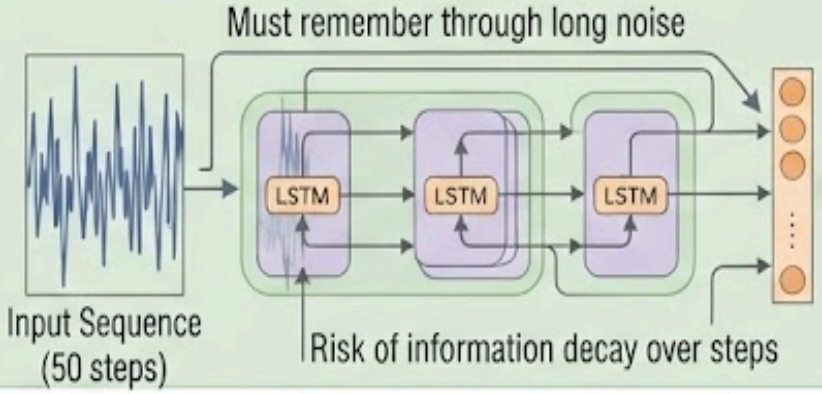
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5. Why the CNN won this task

Burst is local



LSTM struggle with distant memory

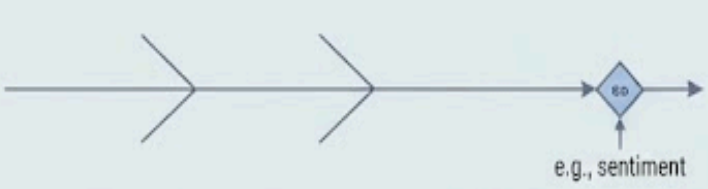


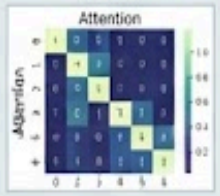
Leopard 1D CNN

Adapting to the Data

Change the task, change the winner

Decision depends on distant past (e.g., sentiment analysis)





Order and distant dependency → Memory/Attention wins

Local patterns in video (e.g., object tracking)



Local spatio-temporal patterns → CNN/ConvLSTM wins

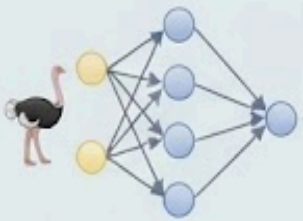
Lesson: Fit the architecture to the data shape, not just to the name.

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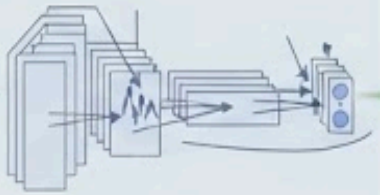
6. Run Full Safari: Where Information is Lost or Kept

Ostrich - FNN / MLP



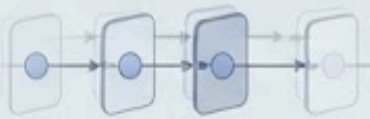
Data moves straight.
Information Kept.

Leopard - CNN



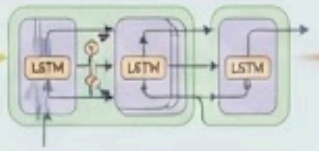
Local patterns found.
Information Kept.

Hyena - RNN



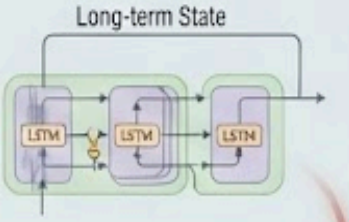
Step-by-step pass.
Some early information fades.

Parrot - LSTM



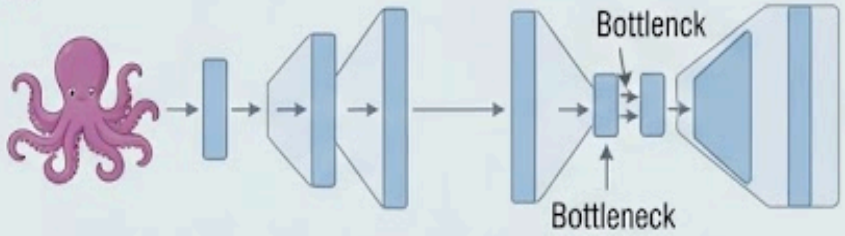
Gates protect states.
Long-term information Kept.

Parrot - LSTM



Gates protect states.
Long-term information Kept.

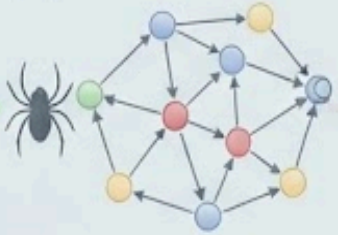
Octopus - Autoencoder



Squeezed through bottleneck. Compression loss.

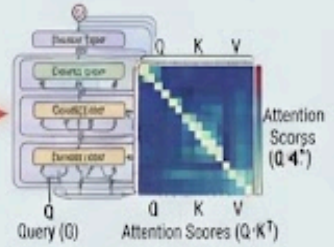
Squeezed through bottleneck.
Compression loss.

Spider - GNN



Messages across web.
Neighborhood information Kept.

Honeybee - Transformer



All-to-all attention.
All sequence context Kept.

Observe how information is either kept pure (FNN, CNN, GNN, Transformer) or intentionally managed and compressed (RNN, LSTM, Autoencoder).

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